Multi-Modal Machine Learning Approach for Threat Detection in WhatsApp and Telegram: Spam Links, Document Based Malware, Chat Behavior, and Sensitive Content Analysis

User behavior and emotions in Telegram and WhatsApp Messages to detect fake accounts.

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Project Proposal Report

B.Sc. (Hons) in Information Technology Specializing in Cyber Security

Sri Lanka Institute of Information Technology Sri Lanka

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

This research proposes a machine learning-driven mobile application to safeguard children from scammers and fake accounts on WhatsApp and Telegram, addressing a critical gap in online safety for vulnerable users in Sri Lanka. The rising use of these encrypted platforms for online learning has heightened exposure to manipulative behaviors, yet existing studies focus on content-level threats, neglecting behavioral and emotional cues. The application captures messages, converts Singlish to English using a fine-tuned Large Language Model (LLM) like BERT, and analyzes emotional tones and behavioral patterns (e.g., message frequency, timing) with ML techniques such as Random Forest and LSTM. It generates real-time threat alerts low, medium, or high risk via an interactive parent dashboard.Projected outcomes include an F1-score exceeding 0.85 in detecting fake accounts, validated with synthetic and public datasets. This novel, region-specific solution enhances child safety, with commercialization potential targeting schools and safety organizations, supported by a 25,500 LKR budget. The study advances multi-modal threat detection, aligning with Sri Lanka's cybersecurity goals.

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List of Abbreviations

Abbreviation		Description
API	-	Application Programming Interface
BERT	-	Bidirectional Encoder Representations from Transformers
LLM	-	Large Language Model
LSTM	-	Long Short-Term Memory
NLP	-	Natural Language Processing
ONNX	-	Open Neural Network Exchange

I. INTRODUCTION

III.1 Background and Literature Review

Messaging platforms such as WhatsApp and Telegram have become integral to daily communication, particularly among young users. While these platforms enable convenient and interactive communication, they are also increasingly exploited by malicious actors who create fake or harmful accounts to spread spam, manipulate users emotionally, or distribute inappropriate content. The detection of such accounts is challenging, as traditional rule-based approaches are insufficient to capture the subtle emotional and behavioral cues present in human communication.

User behavior analysis in messaging applications has been widely studied in the context of spam detection and fraud prevention. Prior research has shown that fake or automated accounts often exhibit distinct behavioral features, such as irregular message timing, high-frequency posting, excessive repetition of similar text, or unusual response delays. These characteristics differentiate them from legitimate users, whose interactions tend to follow more natural social and temporal patterns [1]. In the context of WhatsApp and Telegram, where conversations are mostly private, detecting such anomalies requires careful feature engineering that considers both temporal patterns and interaction dynamics.

Parallel to behavioral analysis, emotion detection in text has gained significant attention in recent years. Machine learning and deep learning methods, such as LSTM, CNN, and Transformer-based models (e.g., BERT), have been successfully applied to identify emotional tones like anger, joy, sadness, or fear in short messages [2]. Emotions play a critical role in threat detection because malicious or fake accounts often attempt to manipulate the emotional state of victims, for instance, by inducing fear, urgency, or curiosity to provoke harmful actions. Several studies on cyberbullying and online harassment also highlight that unusual emotional polarity patterns in chats can serve as indicators of harmful interactions.

Recent work has begun to combine emotional and behavioral cues for more robust fake account detection. For example, hybrid models that integrate linguistic sentiment features with usage patterns have been shown to outperform purely content-based or behavior-based approaches [3]. However, most of these studies focus on platforms like Twitter, Facebook, or WeChat, while research specifically targeting encrypted, closed-group platforms like WhatsApp and Telegram remains limited. This gap is significant in the Sri Lankan context, where children's increasing use of these apps for online learning has heightened exposure to risks such as spam links, predatory chats, and manipulative emotional messaging.

Therefore, analyzing user behavior (timing, frequency, repetition) alongside emotional tone detection in WhatsApp and Telegram conversations provides a promising direction for detecting fake accounts and safeguarding young users. This approach not only advances existing research on multi-modal threat detection but also addresses a critical real-world problem by applying it to platforms that are widely used yet understudied in terms of automated safety solutions.

III.2 Research Gap

While numerous studies have explored spam detection, phishing links, and malware in social media and messaging platforms, there is limited research focused on Telegram and WhatsApp user behavior and emotional patterns. Existing work mainly emphasizes content-level threats, such as malicious links or harmful media, but deceptive user behavior and emotional cues remain underexplored. Most approaches to fake account detection rely on metadata (e.g., profile information or network activity), overlooking the linguistic tone, sentiment, and emotional consistency in conversations.

Furthermore, prior research rarely integrates both behavioral features (message frequency, timing, and repetition) with emotional tone analysis, even though this combination may reveal strong indicators of fake or harmful accounts. Specifically, Telegram and WhatsApp contexts are understudied, despite their increasing use among vulnerable groups such as children during online learning. This creates a critical gap in understanding how real users differ from fake or malicious users in communication style, emotional signals, and activity patterns.

Table I-1 - Research Gap in tabular view

Research Paper	Emotion Analysis	Behavioral Analysis	Combined Emotion + Behavior	Platform Specific (WhatsApp/Telegram)
Research [1]	Implemented	Not Implemented	Not Implemented	Telegram only
Research [2]	Not Implemented	Implemented	Not Implemented	WhatsApp only
Research [3]	Implemented	Implemented	Not Implemented	General Social Media
Proposed Solution	Implemented	Implemented	Implemented	WhatsApp + Telegram

III.3 Research Problem

During the COVID-19 pandemic, the increased use of messaging applications such as WhatsApp and Telegram among school children in Sri Lanka under the age of 16 created both opportunities and risks. While these platforms supported online learning and peer communication, they also became a gateway for scammers and malicious actors to exploit vulnerable users.

Children are particularly at risk because they lack the maturity and awareness to identify suspicious behavior, fake accounts, or harmful conversations. Scammers often send deceptive messages, fake links, or emotionally manipulative texts that can mislead children into:

- Clicking on phishing links that may steal personal information.
- Downloading malicious files disguised as study materials or entertainment.
- Engaging in harmful chats where they can be emotionally manipulated or coerced.

- Being exposed to inappropriate or sensitive content can negatively affect their psychological well-being.

Unlike adults, children may not fully understand the long-term consequences of interacting with suspicious accounts. For example, they may trust a fake identity posing as a classmate or teacher, leading them to share sensitive personal details or follow harmful advice. These manipulative message patterns and emotional cues can result in emotional harm, behavioral changes, and even long-term safety risks.
At present, parents and teachers cannot continuously monitor children's private chat activities due to both practical limitations and privacy concerns. Therefore, there is a pressing need for a smart and user-friendly solution that can automatically detect suspicious behavior and emotional manipulation in text messages. Such a system should alert users or guardians when harmful or deceptive patterns are detected, without invading the child's privacy by storing or exposing entire conversations.

Thus, the core research problem is:
How can emotional and behavioral analysis of text messages be used to detect fake, suspicious, or harmful accounts in messaging platforms like WhatsApp and Telegram, in order to protect children from online scams and emotional manipulation?

II. OBJECTIVES

III.1 Main Objective

To develop a cross-platform mobile application that integrates advanced multi-modal machine learning algorithms, including natural language processing (NLP), computer vision, and behavioral analytics, alongside sandbox-based dynamic analysis techniques, to proactively identify and mitigate a range of cybersecurity threats within popular messaging platforms such as WhatsApp and Telegram. This encompasses the detection of spam and malicious links through simulated URL behavior analysis, identification of document-based malware (e.g., in PDFs, DOCs, and APKs) via runtime execution monitoring, analysis of suspicious chat behaviors and emotional tones to flag fake or harmful accounts based on patterns like message timing, frequency, repetition, and sentiment anomalies, and filtering of sensitive multimedia content (e.g., images and videos containing nudity, violence, or explicit material) using real-time classification models. The overarching goal is to enhance digital safety for vulnerable users, particularly minors in educational and social contexts, by providing intuitive alerts, user-friendly interfaces, and privacy-preserving mechanisms that prevent platform misuse, reduce exposure to online risks, and promote secure communication practices in alignment with national cybersecurity standards in Sri Lanka.

III.2 Sub-Objectives

To investigate and model user behavior and emotional tones in WhatsApp and Telegram messages by leveraging machine learning techniques, including natural language processing for sentiment analysis and feature extraction for behavioral patterns such as message timing, frequency, and repetition, in order to detect and classify fake or suspicious accounts through integrated anomaly detection models, evaluate their performance via accuracy metrics, and provide visual interpretations through interactive

dashboards, ultimately contributing to enhanced threat detection and safer user interactions within the overarching mobile application framework.

III. METHODOLOGY

The methodology develops a mobile application module to protect children from scammers on WhatsApp and Telegram by analyzing messages for fake account detection. Using Python and libraries like Hugging Face Transformers and Scikit-learn, the process includes four stages: Message Capture, Language Processing, ML Analysis, and Alert Generation.

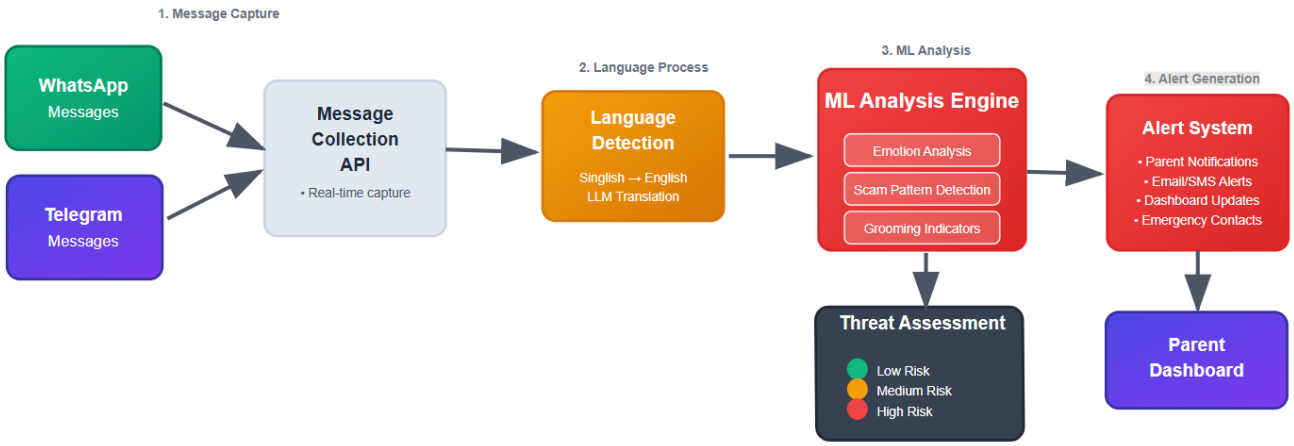

Figure 1 - System Diagram

Message Capture
Messages are collected in real-time from the child's WhatsApp and Telegram accounts using APIs (e.g., WhatsApp Business API, Telegram Bot API) with parental consent, extracting text and metadata from simulated or anonymized datasets for testing.

Language Processing
Messages are checked for Singlish using a language detection tool, converting them to English with a fine-tuned LLM (e.g., BERT) optimized for mobile use via ONNX Runtime, ensuring accurate semantic preservation.

ML Analysis
English messages are analyzed for emotional tones (via BERT sentiment classification) and behavioral patterns (e.g., frequency, timing, repetition) using a Random Forest or LSTM model. Trained on public and synthetic datasets, the model outputs threat levels (low, medium, high) using TensorFlow Lite for real-time, on-device processing.

Alert Generation

Threat levels trigger alerts: medium risk updates a parent dashboard (via Streamlit), high risk sends notifications to guardians. The dashboard visualizes emotional and behavioral insights, with testing to refine alert accuracy and privacy.

This approach ensures a scalable, privacy-compliant solution, validated with stakeholder feedback, to safeguard children in Sri Lanka.

III.1 Requirement Gathering

The requirement gathering process for the development of a machine learning-driven module to analyze user behavior and emotional tones in WhatsApp and Telegram messages for detecting fake or suspicious accounts involved a comprehensive approach to ensure alignment with the project's objectives. This process encompassed the following key activities

Extensive Literature Review: A thorough review of recent academic papers, journal articles, and technical reports on natural language processing (NLP), sentiment analysis, behavioral analytics, and anomaly detection in messaging platforms was conducted. This review focused on understanding current methodologies for detecting fake accounts, limitations of existing systems, and emerging trends in chat-based threat detection, particularly for WhatsApp and Telegram. Insights from this analysis helped identify gaps in real-time behavioral and emotional profiling, guiding the design of a novel, platform-specific solution.

Stakeholder Consultation: Engagements with cybersecurity experts, child safety advocates, and mobile application developers were carried out through interviews, surveys, and workshops. These consultations provided practical perspectives on the challenges of identifying deceptive accounts in messaging apps, user expectations for privacy, and the need for lightweight, real-time detection systems suitable for mobile devices. Feedback from stakeholders, including insights from Sri Lanka CERT|CC and child protection agencies, ensured the module addresses real-world safety concerns for vulnerable users, such as minors.

System Requirements Analysis: Detailed technical specifications were documented, focusing on the module's requirements for NLP-based sentiment analysis, behavioral feature extraction (e.g., message timing, frequency, and repetition), and machine learning model performance. This analysis also covered scalability for real-time processing on mobile devices, privacy-preserving techniques for encrypted platforms, and integration with the broader application's user interface to deliver intuitive alerts and dashboards.

Use Case Scenarios: Multiple use case scenarios were developed to model how the module will function in diverse WhatsApp and Telegram environments. These scenarios included identifying fake accounts spreading spam, detecting manipulative behavior in group chats, and flagging emotionally harmful interactions. By simulating real-world communication patterns, these use cases ensured the module addresses practical challenges and delivers effective, user-centric threat detection in dynamic messaging contexts. Past Research Analysis

III.2.2-i Literature Review of Key Studies:

The following key studies provide foundational insights into the use of natural language processing (NLP), sentiment analysis, and behavioral analytics for detecting suspicious or fake accounts in social media and messaging platforms, with a focus on their relevance to the proposed module for WhatsApp and Telegram.

Research [4]: Mansoori et al. (2020)
This study explores sentiment analysis using NLP to detect suspicious activities on Twitter and Facebook, employing a Long Short-Term Memory (LSTM) network to analyze emotional tones in social media posts. The authors emphasize the effectiveness of LSTM in capturing long-term dependencies in text, achieving high accuracy in identifying malicious behavioral profiles. However, the study focuses on public social media platforms and does not address the unique challenges of encrypted messaging apps like WhatsApp and Telegram, such as limited data access and real-time processing constraints. Additionally, it lacks integration of behavioral features like message timing or frequency, which are critical for identifying fake accounts in chat-based environments

Research [5]: Gagandeep and Verma (2024)
This work proposes an anomaly detection system for social media posts using NLP and behavioral profiling to flag suspicious activities. The methodology combines sentiment analysis with heuristics to detect malicious behavior, achieving improved precision and recall compared to traditional approaches. While the study highlights the importance of behavioral features, it primarily focuses on open platforms like X and does not address the specific context of private messaging apps. The lack of focus on real-time, mobile-optimized detection and the absence of emotional tone analysis tailored to encrypted platforms limit its applicability to WhatsApp and Telegram.

Research [6]: Ahmed et al. (2023)
This research investigates sentiment analysis and machine learning for news classification and fake news detection across digital platforms, achieving 89% accuracy in sentiment classification. It discusses the use of NLP to analyze textual content for deceptive patterns but does not specifically target messaging apps like WhatsApp or Telegram. The study also overlooks behavioral analytics, such as message timing or repetition, and does not address the challenges of lightweight model deployment on mobile devices, which are essential for real-time threat detection in the proposed module.

These studies collectively highlight the potential of NLP and sentiment analysis for detecting suspicious activities but fall short in addressing the specific requirements of WhatsApp and Telegram, such as encrypted data processing, real-time behavioral profiling, and mobile-optimized performance. The proposed module aims to bridge these gaps by integrating emotional and behavioral analytics tailored to messaging platforms, offering a novel approach to fake account detection.

III.2.2-ii Identification of Gaps

The literature on sentiment analysis and anomaly detection for social media and messaging platforms reveals critical shortcomings that the proposed module, focused on analyzing user behavior and emotional tones in WhatsApp and Telegram to detect fake accounts, aims to address. The following gaps are identified:

Real-Time Emotional and Behavioral Analysis in Encrypted Messaging Platforms: Existing research primarily focuses on sentiment analysis for open social media platforms, neglecting the unique challenges of encrypted messaging apps like WhatsApp and Telegram. These platforms restrict data access and require real-time processing for effective threat detection on mobile devices, an area underexplored in current studies. The proposed module seeks to address this by developing real-time emotional and behavioral analytics tailored to these encrypted environments.

Integration of Behavioral Features like Message Timing and Frequency: Most studies rely heavily on textual content for detecting suspicious activities, overlooking behavioral indicators such as message timing, frequency, or repetition. These features are essential for identifying fake or malicious accounts in messaging apps, where patterns like rapid messaging or activity at unusual hours can signal deceptive behavior. The proposed module aims to fill this gap by combining emotional tone analysis with temporal and frequency-based behavioral profiling to improve detection accuracy.

Mobile-Optimized and Privacy-Preserving Detection Systems: Current research often fails to address the need for lightweight, mobile-optimized models that can perform real-time detection on resource-constrained devices. Additionally, there is limited focus on privacy-preserving techniques for analyzing encrypted messages, particularly for protecting vulnerable users like minors. The proposed module tackles this gap by designing a lightweight, privacy-aware system that ensures efficient performance on mobile devices while adhering to ethical standards and national cybersecurity guidelines in Sri Lanka.

These gaps underscore the need for a novel approach that integrates real-time emotional and behavioral analytics, specifically designed for encrypted messaging platforms, to effectively detect fake accounts while prioritizing usability and privacy.

III.2 Feasibility Study

The technical feasibility of the proposed research component involves evaluating the appropriateness of the tools, technologies, and techniques required to develop a machine learning-driven module for analyzing user behavior and emotional tones in WhatsApp and Telegram messages to detect fake or suspicious accounts. This module will process messages, including converting Singlish to English using a Large Language Model (LLM), analyze emotional and behavioral patterns, and issue real-time threat alerts. The feasibility is addressed in the following areas:

III.3.1 Technical Feasibility

III.2.2-i Tools and Technology

- **Python**: This versatile, general-purpose programming language will be used for developing core functionalities of the module, including data preprocessing, feature extraction, and machine learning model implementation. Python's extensive libraries, such as TensorFlow, PyTorch, and NLTK, support natural language processing (NLP) and behavioral analytics, making it ideal for building and testing sentiment analysis and anomaly detection models.
- **Jupyter Notebook**: This interactive development environment will be employed for prototyping, testing, and visualizing machine learning models. Jupyter Notebook facilitates exploratory data analysis, model training, and dashboard creation, enabling iterative development of the module's emotional and behavioral profiling capabilities.
- **Streamlit**: This open-source framework will be utilized to develop interactive dashboards for visualizing detection results. Streamlit allows for the creation of user-friendly interfaces to display real-time insights, such as suspicious account alerts and behavioral pattern visualizations, enhancing the module's usability on mobile devices.

III.2.2-ii Algorithm and Architectures

- **BERT (Bidirectional Encoder Representations from Transformers)**: This LLM will be used for both Singlish-to-English conversion and sentiment analysis of messages. BERT's contextual understanding of text makes it ideal for detecting emotional tones (e.g., positive, negative, manipulative) in WhatsApp and Telegram chats, ensuring accurate identification of suspicious patterns.

- **Random Forest**: This ensemble machine learning algorithm will be utilized for anomaly detection, combining emotional and behavioral features (e.g., message timing, frequency, repetition) to classify user profiles as legitimate or suspicious. Its robustness and efficiency suit real-time detection on resource-constrained mobile devices.

- **TF-IDF (Term Frequency-Inverse Document Frequency)**: This text processing technique will be applied to extract textual features, such as repetitive phrases or keywords, from messages to

support behavioral profiling. TF-IDF complements BERT by providing lightweight feature extraction for anomaly detection.

III.2.2-iii Techniques

- Singlish-to-English Conversion (LLM): A fine-tuned BERT model will be used to detect and translate Singlish messages into standard English, ensuring consistency for subsequent emotional and behavioral analysis. This technique addresses the linguistic diversity of Sri Lankan users, enabling accurate processing of colloquial messages.

- Sentiment and Behavioral Analysis: NLP techniques, leveraging BERT for sentiment classification and statistical methods for extracting behavioral features (e.g., message frequency, timing patterns), will be implemented to identify fake accounts. These techniques enable real-time detection of manipulative or anomalous behaviors in chat environments.

- Performance Evaluation and Visualization: The module's performance will be assessed using metrics like accuracy, precision, recall, and F1-score to ensure reliable detection of fake accounts. Interactive dashboards, built with Plotly Dash, will visualize emotional and behavioral patterns, providing intuitive alerts to users about potential threats while maintaining privacy compliance.

III.3 PROJECT REQUIREMENTS

The project requirements for the machine learning-driven module to analyze user behavior and emotional tones in WhatsApp and Telegram messages for detecting fake or suspicious accounts are outlined below. These requirements ensure the module effectively processes messages, including converting Singlish to English, performs real-time emotional and behavioral analysis, and delivers user-friendly threat alerts, aligning with the project's goal of enhancing user safety in messaging platforms.

III.3.1 Functional Requirements

- **Message Processing and Singlish Conversion**: The system must retrieve and process messages from WhatsApp and Telegram, identifying and converting Singlish (a colloquial mix of English, Sinhala, Tamil, and other influences) to standard English using a Large Language Model (e.g., BERT or a fine-tuned GPT model). This ensures consistent input for subsequent emotional and behavioral analysis, preserving semantic accuracy for Sri Lankan users.

- **Emotional and Behavioral Analysis**: The module must analyze messages to extract emotional tones (e.g., positive, negative, manipulative) using natural language processing (NLP) techniques and behavioral features (e.g., message timing, frequency, repetition) to build user profiles. Machine learning models, such as BERT for sentiment analysis and Random Forest for anomaly detection, will classify accounts as legitimate or suspicious based on these patterns.

- **Threat Alert Generation**: The system must generate real-time, user-friendly alerts (e.g., visual notifications or warnings) through the mobile application's interface to inform users of potential fake or malicious accounts. Alerts will be tiered (e.g., low, medium, high risk) based on model confidence scores, enabling intuitive user interaction without compromising privacy.

III.3.2 Non-Functional Requirements

- **Accuracy**: The module must achieve high accuracy in detecting fake or suspicious accounts, correctly identifying deceptive patterns while minimizing false positives. This ensures reliable differentiation between legitimate and malicious users based on emotional and behavioral analysis.

- **Performance**: The system must provide low-latency, real-time processing of messages to support seamless operation in dynamic chat environments. Models must be optimized for mobile devices, ensuring efficient performance without excessive computational overhead.

- **Availability**: The module must maintain high availability, operating reliably across diverse network conditions and handling high volumes of messages without degradation in performance, ensuring consistent threat detection for users.

- **Privacy and Security**: The system must adhere to strict privacy standards, processing encrypted messages in a privacy-preserving manner, especially for vulnerable users like minors. It must comply with national cybersecurity guidelines in Sri Lanka, preventing unauthorized access to user data and ensuring ethical handling of sensitive information.

III.4 WORK BREAKDOWN AND GRANT CHART

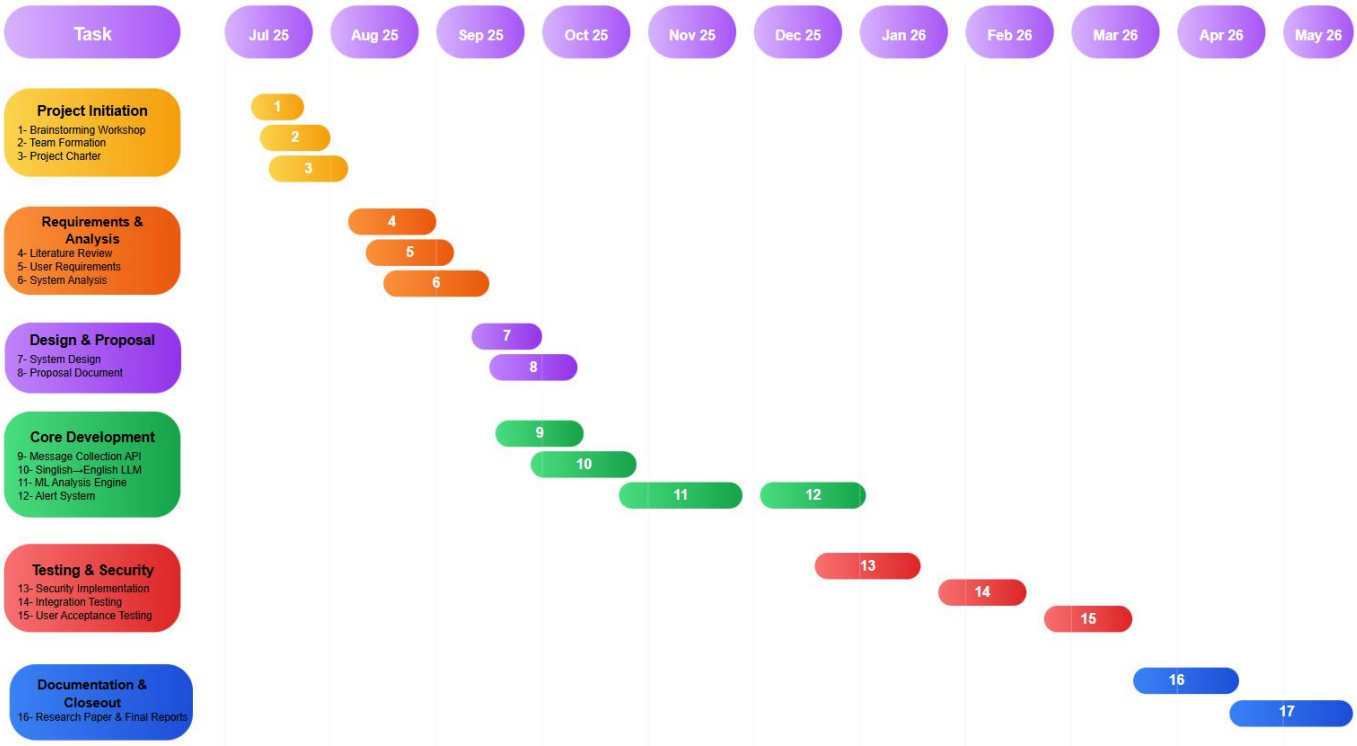

Figure 2 – Grant chart

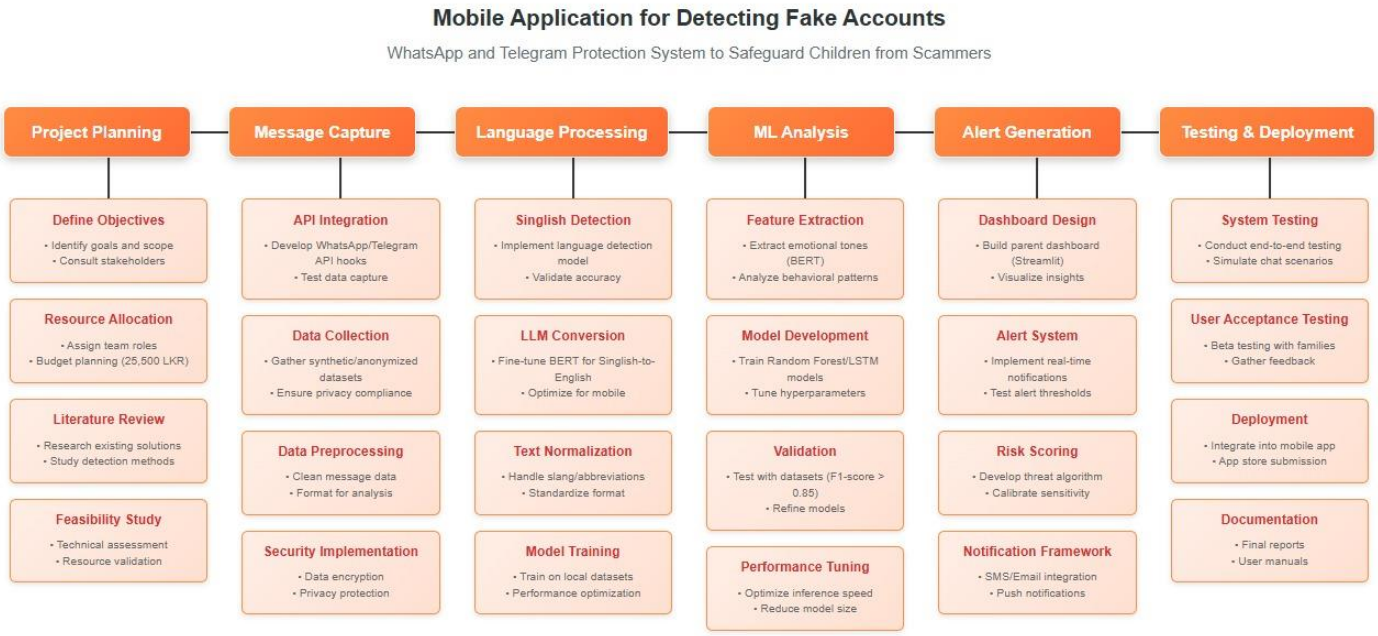

Figure 3 - Work Breakdown Chart

III.5 COMMERCIALIZATION

The proposed machine learning-driven module for analyzing user behavior and emotional tones in WhatsApp and Telegram messages to detect fake or suspicious accounts has significant potential for commercialization within the growing market of cybersecurity and child online safety solutions. By integrating advanced features such as Singlish-to-English conversion, real-time emotional and behavioral analysis, and user-friendly threat alerts into a mobile application, the solution offers a robust tool for enhancing user safety on popular messaging platforms. The application will allow users, particularly parents, educators, and minors, to monitor chat interactions, receive real-time alerts about potential fake accounts, and visualize behavioral patterns through interactive dashboards, providing a seamless and intuitive experience.

The commercialization strategy targets educational institutions, child safety organizations, and individual users in Sri Lanka and beyond, where messaging apps like WhatsApp and Telegram are widely used for communication and learning. The module's ability to address region-specific linguistic challenges (e.g., Singlish) and its focus on protecting vulnerable users make it highly relevant in markets with diverse linguistic and cultural contexts. Key advantages include its lightweight, mobile-optimized design and privacy-preserving approach, which ensure accessibility and compliance with data protection standards.

Key Commercialization Points:

- **Target Market**: Educational institutions, child safety organizations, cybersecurity providers, and individual users (e.g., parents and minors) seeking safe online communication solutions.

- **Unique Selling Proposition:** The integration of Singlish-to-English conversion, real-time emotional and behavioral analysis, and interactive dashboards for fake account detection offers a unique, region-specific cybersecurity solution for messaging platforms, distinguishing it from generic social media monitoring tools.

- **Revenue Model:** The mobile application can be offered as a freemium service, with basic features available for free and premium features (e.g., advanced analytics, customizable alerts) accessible via subscription fees. Additional revenue can be generated through partnerships with educational platforms or cybersecurity firms integrating the module into their services.

- **Market Entry Strategy**: Initial market entry will focus on Sri Lankan schools, universities, and child protection agencies, leveraging partnerships with organizations like Sri Lanka CERT|CC. Demonstrating the module's effectiveness in protecting minors from online threats will drive adoption, with potential expansion to global markets with similar messaging app usage patterns. Given the increasing demand for safe online communication, particularly for minors, and the module's innovative approach to region-specific threat detection, it holds strong potential for successful commercialization.

IV. BUDGET AND BUDGET JUSTIFICATION

As the proposed module is software-based, no hardware components are required for its implementation. The primary costs are associated with software development, cloud-based computing resources for model training and testing, and publication expenses. The budget also accounts for miscellaneous expenses such as internet connectivity and stationery for documentation. The following table outlines the estimated costs for developing and implementing the module.

Table IV-1 - Budget Analysis

Type	Cost
Internet and Cloud Services	6000 LKR
Publication Cost	15000 LKR
Stationary	4500 LKR
Total	25000 LKR

Budget Justification:

- Internet and Cloud Services (6,000 LKR): Covers subscription fees for cloud platforms (e.g., Google Colab, AWS, or Hugging Face) used for training and testing machine learning models, including LLMs for Singlish conversion and ML models for emotional and behavioral analysis. This ensures access to sufficient computational resources for processing large datasets and optimizing models.
- Publication Cost (15,000 LKR): Accounts for expenses related to submitting the research to academic journals, conferences, or open-access repositories to disseminate findings and gain visibility for potential commercialization.
- Stationery (4,500 LKR): Includes costs for printing, documentation, and other materials needed for project planning, stakeholder consultations, and preparing the final research report.These costs ensure the successful development, evaluation, and dissemination of the module, supporting its integration into a broader mobile application for real-time fake account detection.

Summary

This research proposes a mobile application to protect children from scammers on WhatsApp and Telegram by detecting fake accounts. It captures messages, converts Singlish to English using a Large Language Model (LLM), and analyzes emotional tones and behavioral patterns (e.g., frequency, timing) with machine learning (ML) techniques like BERT and Random Forest, issuing real-time threat alerts via a parent dashboard. The methodology includes message capture, language processing, ML analysis, and alert generation, using Python and optimized models for mobile use. Addressing a gap in Telegram/WhatsApp safety research, especially for children's online learning, the project is feasible with tools like Hugging Face Transformers and Plotly Dash. Commercialization targets schools and child safety organizations with a freemium model, supported by a 25,500 LKR budget for cloud services and publication, aligning with Sri Lanka's cybersecurity goals.

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